

1) A 3x3 filter and a 5x5 grayscale image are provided. Calculate the size of the output feature map and describe the convolution procedure. Additionally, include examples to illustrate how stride and padding affect output size.

2) You attempted transfer learning for satellite picture classification using a VGG16 model pretrained on ImageNet, however your accuracy was subpar. Examine the possible causes of this and the actions you can take to improve the efficacy of transfer learning in this area.

3) You have been tasked with creating an AI system that can categorize brief video clips into groups like "running," "dancing," and "playing guitar." Every clip is a collection of thirty grayscale image frames.

Explain how you would create a model to address this issue by combining Long Short-Term Memory (LSTM) with Convolutional Neural Networks (CNN).

CNN would be used in what aspect of the model, and why?

After CNN, how would LSTM be applied?

Describe the steps involved in data pretreatment and input shape.

What are the advantages of employing CNN and LSTM together as opposed to just one of them?

4) You are developing a self-driving automobile AI system. Describe the use of CNNs in the system's at least three vision-related tasks. Additionally, explain how CNNs perform well in certain real-time applications thanks to spatial properties.

5) In order to predict the next word in lengthy English sentences, you are developing a system. Describe how an LSTM preserves long-term dependencies while assisting in the solution of the vanishing gradient problem. Explain the three LSTM gates and their purposes.

6) You are creating a voice assistant with two functions:

Determine the speaker's emotion (happy, sad, furious, etc.) by transcribing their words to text.

Create a recurrent neural network design with distinct RNN topologies to manage both jobs.

Which topology—Many-to-Many, Many-to-One, etc.—would you employ for each activity, and why?

Can the same RNN encoder be used for both tasks?

What difficulties may such a multi-task RNN present during training, and how would you overcome them?